

# Variables & Equations

ProMo

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# 1 Variables

## 2 root

|    | var                | symbol      | documentation                                | type    | units | eqs |
|----|--------------------|-------------|----------------------------------------------|---------|-------|-----|
| 10 | $S_{I,q}$          | S_Iq        | selection matrix interface to control output | network |       |     |
| 8  | $F^{sink}_{N,A}$   | F_NA_sink   | incidence matrix NA sink                     | network |       |     |
| 7  | $F^{source}_{N,A}$ | F_NA_source | incidence matrix NA source                   | network |       |     |
| 20 | $A_{N,t,u}$        | A_Ntu       | mapping from input elements to outputs       | network |       |     |
| 17 | $cz_N$             | cz_N        | output from control                          | network |       |     |
| 3  | $F^{source}_{N,I}$ | F_NI_source | incidence matrix NI source                   | network |       |     |
| 21 | $u_{N,t,u}$        | u_Ntu       | input signal in control domain               | network |       |     |
| 4  | $F^{sink}_{N,I}$   | F_NI_sink   | incidence matrix NI sink                     | network |       |     |
| 27 | $I_{N,A}$          | I_NA        | identity mapping from <N> to <A>             | network |       |     |
| 2  | $F_{N,A}$          | F           | incidence matrix                             | network |       |     |
| 11 | $I_{t,u}$          | I_tu        | identity mapping from <t> to <u>             | network |       |     |
| 13 | $S_{I,q}$          | S_Aq        | selection matrix arcs to outputs             | network |       |     |

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|     | var                | symbol      | documentation                                       | type    | units | eqs |
|-----|--------------------|-------------|-----------------------------------------------------|---------|-------|-----|
| 6   | $F^{sink}_{A,I}$   | F_AI_sink   | incidence matrix AI sink                            | network |       |     |
| 16  | $mv_I$             | mv_I        | interface variable macro → control                  | network |       |     |
| 22  | $y_{N,t,u}$        | y_Ntu       | output signal in control domain                     | network |       |     |
| 19  | $A_{N,p,q}$        | A_Npq       | mapping from inputs to outputs                      | network |       |     |
| 14  | $S_{N,p,q}$        | S_Npu       | selection matrix for stacker                        | network |       |     |
| 18  | $cz_I$             | cz_I        | interface variable macro → control                  | network |       |     |
| 9   | $S_{I,p}$          | S_Ip        | selection matrix interface to control input         | network |       |     |
| 15  | $S_{N,q,t}$        | S_Nqt       | selection matrix or splitter                        | network |       |     |
| 12  | $S_{A,p}$          | S_Ap        | selection matrix interface species-related measures | network |       |     |
| 5   | $F^{source}_{A,I}$ | F_AI_source | incidence matrix AI source                          | network |       |     |
| 1   | $t$                | t           | time                                                | frame   | s     |     |
| 107 | $\Delta t$         | t_interval  | time interval                                       | frame   | s     | 6   |
| 105 | $t^o$              | to          | starting time                                       | frame   | s     | 4   |
| 106 | $t^e$              | te          | end time                                            | frame   | s     | 5   |

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|     | var | symbol         | documentation            | type     | units | eqs |
|-----|-----|----------------|--------------------------|----------|-------|-----|
| 103 | 1   | <b>one</b>     | numerical value one      | constant |       | 2   |
| 101 | #   | <b>value</b>   | numerical value          | constant |       |     |
| 104 | 0.5 | <b>oneHalf</b> | numerical value one half | constant |       | 3   |
| 102 | 0   | <b>zero</b>    | numerical value zero     | constant |       | 1   |

### 3 physical

|     | var         | symbol       | documentation                        | type     | units                               | eqs |
|-----|-------------|--------------|--------------------------------------|----------|-------------------------------------|-----|
| 24  | $r_{yN}$    | <b>r_y</b>   | y-coordinate                         | frame    | $m$                                 |     |
| 25  | $r_{zN}$    | <b>r_z</b>   | z-coordinate                         | frame    | $m$                                 |     |
| 23  | $r_{xN}$    | <b>r_x</b>   | x-coordinate                         | frame    | $m$                                 |     |
| 144 | $C_N$       | <b>C</b>     | fundamental state – charge           | state    | $A\,s$                              |     |
| 111 | $n_{N,S}$   | <b>n</b>     | fundamental state – molar mass       | state    | $mol$                               | 93  |
| 109 | $S_N$       | <b>S</b>     | fundamental state – internal entropy | state    | $kg\,m^2\,K^{-1}\,s^{-2}$           |     |
| 108 | $U_N$       | <b>U</b>     | fundamental state – internal energy  | state    | $kg\,m^2\,s^{-2}$                   |     |
| 110 | $V_N$       | <b>V</b>     | volume                               | state    | $m^3$                               | 7   |
| 137 | $m_N$       | <b>m</b>     | mass                                 | state    | $kg$                                | 30  |
| 132 | $\lambda_S$ | <b>Mm</b>    | molecular masses                     | constant | $kg\,mol^{-1}$                      |     |
| 121 | $N^A$       | <b>Avo</b>   | Avogadro constant                    | constant | $mol^{-1}$                          |     |
| 122 | $k^B$       | <b>Boltz</b> | Boltzmann constant                   | constant | $kg\,m^2\,K^{-1}\,s^{-2}$           |     |
| 123 | $R$         | <b>R</b>     | gas constant                         | constant | $kg\,m^2\,mol^{-1}\,K^{-1}\,s^{-2}$ | 17  |

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|     | var                                             | symbol     | documentation           | type           | units        | eqs                |
|-----|-------------------------------------------------|------------|-------------------------|----------------|--------------|--------------------|
| 150 | $A_{yzN}$<br>$\rho_N$<br>$A_{xzN}$<br>$A_{xyN}$ | <b>Ayz</b> | cross sectional area yz | secondaryState | $m^2$        | <a href="#">42</a> |
| 143 |                                                 | <b>rho</b> | density                 | secondaryState | $kg\,m^{-3}$ | <a href="#">36</a> |
| 149 |                                                 | <b>Axz</b> | cross sectional are xz  | secondaryState | $m^2$        | <a href="#">41</a> |
| 148 |                                                 | <b>Axy</b> | cross sectional area xy | secondaryState | $m^2$        | <a href="#">40</a> |

## 4 macroscopic

|     | var                | symbol | documentation                                                        | type      | units             | eqs   |
|-----|--------------------|--------|----------------------------------------------------------------------|-----------|-------------------|-------|
| 154 | $\hat{n}_{xA,S}^d$ | fnd_x  | diffusion flow in x-direction                                        | transport | $mol\ s^{-1}$     | 46 89 |
| 194 | $\dot{n}_{xN,S}^c$ | anc_x  | accumulation of molar mass due to convection                         | transport | $mol\ s^{-1}$     | 87    |
| 204 | $\dot{H}_{xN}^c$   | aHnc_x | accumulation of enthalpy due to convective mass flow in x-direction  | transport | $kg\ m^2\ s^{-3}$ | 99    |
| 160 | $\hat{n}_{xA,S}^c$ | fnc_x  | molar convective flow in x-direction                                 | transport | $mol\ s^{-1}$     | 52    |
| 155 | $\hat{n}_{yA,S}^d$ | fnd_y  | diffusion flow in y-direction                                        | transport | $mol\ s^{-1}$     | 47 90 |
| 158 | $c_{A,S}$          | c_AS   | concentration in convective event-dynamic flow                       | transport | $m^{-3}\ mol$     | 50    |
| 208 | $\dot{q}_{xN}$     | aq_x   | accumulation due to heat flow in x-direction                         | transport | $kg\ m^2\ s^{-3}$ | 103   |
| 151 | $\hat{q}_{xA}$     | fq_x   | heat flow in x-direction                                             | transport | $kg\ m^2\ s^{-3}$ | 43    |
| 195 | $\dot{n}_{xN,S}^d$ | and_x  | accumulation due to diffusion in x-direction                         | transport | $mol\ s^{-1}$     | 88    |
| 156 | $\hat{n}_{zA,S}^d$ | fnd_z  | diffusion flow in z-direction                                        | transport | $mol\ s^{-1}$     | 48 91 |
| 210 | $\dot{q}_{zN}$     | aq_z   | accumulation due to heat flow in z-direction                         | transport | $kg\ m^2\ s^{-3}$ | 105   |
| 207 | $\dot{H}_{zN}^d$   | aHnd_z | accumulation of enthalpy due to diffusional mass flow in z-direction | transport | $kg\ m^2\ s^{-3}$ | 102   |
| 212 | $\dot{n}_{yN,S}^d$ | and_y  | accumulation due to diffusion in y-direction                         | transport | $mol\ s^{-1}$     | 107   |

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|     | var                        | symbol    | documentation                                                        | type       | units                | eqs |
|-----|----------------------------|-----------|----------------------------------------------------------------------|------------|----------------------|-----|
| 213 | $\dot{n}_{zN,S}^d$         | and_z     | accumulation due to diffusion in z-direction                         | transport  | $mol\ s^{-1}$        | 108 |
| 209 | $\dot{q}_{yN}$             | aq_y      | accumulation due to heat flow in y-direction                         | transport  | $kg\ m^2\ s^{-3}$    | 104 |
| 152 | $\hat{q}_{yA}$             | fq_y      | heat flow in y-direction                                             | transport  | $kg\ m^2\ s^{-3}$    | 44  |
| 159 | $\hat{V}_A$                | fV        | volumetric flow in x-direction                                       | transport  | $m^3\ s^{-1}$        | 51  |
| 234 | $\hat{m}_{N,A}$            | fm        | convective mass flow                                                 | transport  | $kg\ s^{-1}$         | 133 |
| 157 | $d_A$                      | d         | flow direction of convective flow                                    | transport  |                      | 49  |
| 153 | $\hat{q}_{zA}$             | fq_z      | heat flow in z-direction                                             | transport  | $kg\ m^2\ s^{-3}$    | 45  |
| 211 | $\hat{w}_A$                | fw        | a fixed work flow to start with                                      | transport  | $kg\ m^2\ s^{-3}$    | 106 |
| 205 | $\dot{H}_{xN}^d$           | aHnd_x    | accumulation of enthalpy due to diffusional mass flow in x-direction | transport  | $kg\ m^2\ s^{-3}$    | 100 |
| 214 | $\dot{w}_N$                | aw        | accumulation of enthalpy due to work flow                            | transport  | $kg\ m^2\ s^{-3}$    | 109 |
| 206 | $\dot{H}_{yN}^d$           | aHnd_y    | accumulation of enthalpy due to diffusional mass flow in y-direction | transport  | $kg\ m^2\ s^{-3}$    | 101 |
| 186 | $k_{xA}^q$                 | kqA_x     | thermal conductivity in arc and x-direction                          | properties | $kg\ K^{-1}\ s^{-3}$ | 79  |
| 190 | $\hat{k}_x^{d,Fick_{A,S}}$ | kdAFick_x | Fick's diffusivity in arc and x-direction                            | properties | $m\ s^{-1}$          | 83  |
| 189 | $\rho_A$                   | rhoA      | density in arc                                                       | properties | $kg\ m^{-3}$         | 82  |

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|     | var                        | symbol      | documentation                                       | type       | units                    | eqs       |
|-----|----------------------------|-------------|-----------------------------------------------------|------------|--------------------------|-----------|
| 185 | $k_{zA}^c$                 | kcA_z       | convective mass conductivity in arc and y-direction | properties | $m^{-1} s$               | 78        |
| 184 | $k_{yA}^c$                 | kcA_y       | convective mass conductivity in arc and y-direction | properties | $m^{-1} s$               | 77        |
| 180 | $k_{xA,S}^d$               | kdA_x       | diffusivity in arc and x-direction                  | properties | $kg^{-1} m^{-4} mol^2 s$ | 73        |
| 219 | $R_N^e$                    | elResistant | electrical resistant                                | properties | $kg m^2 A^{-2} s^{-3}$   | 115       |
| 187 | $k_{yA}^q$                 | kqA_y       | thermal conductivity in arc and y-direction         | properties | $kg K^{-1} s^{-3}$       | 80        |
| 192 | $\hat{k}_{z,A,S}^{d,Fick}$ | kdAFick_z   | Fick diffusivity in arc and z-direction             | properties | $ms^{-1}$                | 85        |
| 191 | $\hat{k}_{y,A,S}^{d,Fick}$ | kdAFick_y   | Fick diffusivity in arc and y-direction             | properties | $ms^{-1}$                | 84        |
| 181 | $k_{yA,S}^d$               | kdA_y       | diffusivity in arc and y-direction                  | properties | $kg^{-1} m^{-4} mol^2 s$ | 74        |
| 188 | $k_{zA}^q$                 | kqA_z       | thermal conductivity in arc and z-direction         | properties | $kg K^{-1} s^{-3}$       | 81        |
| 182 | $k_{zA,S}^d$               | kdA_z       | diffusivity in arc and z-direction                  | properties | $kg^{-1} m^{-4} mol^2 s$ | 75        |
| 183 | $k_{xA}^c$                 | kcA_x       | convective mass conductivity in arc and x diretion  | properties | $m^{-1} s$               | 76        |
| 193 | $h_{A,S}$                  | hA          | partial molar enthalpiies in arc                    | properties | $kg m^2 mol^{-1} s^{-2}$ | 86        |
| 115 | $H_N$                      | H           | Enthalpy                                            | state      | $kg m^2 s^{-2}$          | 11<br>112 |
| 116 | $A_N$                      | A           | Helmholtz energy                                    | state      | $kg m^2 s^{-2}$          | 12        |

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|     | var           | symbol          | documentation                               | type           | units                       | eqs       |
|-----|---------------|-----------------|---------------------------------------------|----------------|-----------------------------|-----------|
| 216 | $H^o_N$       | Ho              | initial enthalpy                            | state          | $kg\,m^2\,s^{-2}$           | 111       |
| 203 | $n^o_{N,S}$   | no              | initial mass                                | state          | $mol$                       | 98        |
| 117 | $G_N$         | G               | Gibbs free energy                           | state          | $kg\,m^2\,s^{-2}$           | 13        |
| 161 | $\mu^o_{N,S}$ | chemPotStandard | instantiating standard chemical potential   | effort         | $kg\,m^2\,mol^{-1}\,s^{-2}$ | 53        |
| 217 | $U^e_N$       | Ue              | electrical potential – voltage              | effort         | $kg\,m^2\,A^{-1}\,s^{-3}$   | 113       |
| 112 | $p_N$         | p               | thermodynamic pressure                      | effort         | $kg\,m^{-1}\,s^{-2}$        | 8         |
| 114 | $\mu_{N,S}$   | chemPot         | chemical potential                          | effort         | $kg\,m^2\,mol^{-1}\,s^{-2}$ | 10 54     |
| 113 | $T_N$         | T               | temperature                                 | effort         | $K$                         | 9 121     |
| 124 | $C_{pN}$      | Cp              | total heat capacity at constant pressure    | secondaryState | $kg\,m^2\,K^{-1}\,s^{-2}$   | 18<br>117 |
| 118 | $v_{xN}$      | v_x             | velocity in x-direction                     | secondaryState | $ms^{-1}$                   | 14        |
| 141 | $c_{pN}$      | cp              | specific heat capacity at constant pressure | secondaryState | $m^2\,K^{-1}\,s^{-2}$       | 34<br>120 |
| 222 | $T^{ref}_N$   | T_ref           | reference temperature                       | secondaryState | $K$                         | 119       |
| 142 | $c_{VN}$      | cV              | specific heat capacity at constant volume   | secondaryState | $m^2\,K^{-1}\,s^{-2}$       | 35        |
| 140 | $x_{N,S}$     | x               | mole fraction                               | secondaryState |                             | 33        |

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|     | var               | symbol                | documentation                              | type              | units                       | eqs                 |
|-----|-------------------|-----------------------|--------------------------------------------|-------------------|-----------------------------|---------------------|
| 125 | $C_{VN}$          | <b>CV</b>             | total heat capacity at constant volume     | secondaryState    | $kg\,m^2\,K^{-1}\,s^{-2}$   | <a href="#">19</a>  |
| 120 | $v_{zN}$          | <b>v_z</b>            | velocity in z-direction                    | secondaryState    | $ms^{-1}$                   | <a href="#">16</a>  |
| 136 | $h_{N,S}$         | <b>h</b>              | partial molar enthalpies                   | secondaryState    | $kg\,m^2\,mol^{-1}\,s^{-2}$ | <a href="#">29</a>  |
| 138 | $c_{N,S}$         | <b>c</b>              | molar concentration                        | secondaryState    | $m^{-3}\,mol$               | <a href="#">31</a>  |
| 139 | $n^t_N$           | <b>nt</b>             | total number of moles                      | secondaryState    | $mol$                       | <a href="#">32</a>  |
| 119 | $v_{yN}$          | <b>v_y</b>            | velocity in y-direction                    | secondaryState    | $ms^{-1}$                   | <a href="#">15</a>  |
| 202 | $\tilde{n}_{N,S}$ | <b>np</b>             | link variable np to interface macroscopic  | conversion        | $m^{-3}\,mol\,s^{-1}$       | <a href="#">97</a>  |
| 220 | $\dot{U}^e_A$     | <b>dUe</b>            | Kirkhoffs first law                        | diffState         | $kg\,m^2\,A^{-1}\,s^{-3}$   | <a href="#">116</a> |
| 215 | $\dot{H}_N$       | <b>dH</b>             | accumulation of enthalpy                   | diffState         | $kg\,m^2\,s^{-3}$           | <a href="#">110</a> |
| 196 | $\dot{n}_{N,S}$   | <b>an</b>             | differential mass balance without reaction | diffState         | $mol\,s^{-1}$               | <a href="#">92</a>  |
| 218 | $I^e_N$           | <b>current</b>        | current definition                         | internalTransport | $A$                         | <a href="#">114</a> |
| 224 | $\bar{T}_N$       | <b>T_meas</b>         | temperature measurement                    | observation       |                             | <a href="#">123</a> |
| 223 | $T^n_N$           | <b>T_meas_norming</b> | value to norm measurement of temperature   | observation       | $K$                         | <a href="#">122</a> |

## 5 reactions

|     | var                 | symbol             | documentation                          | type           | units                       | eqs                |
|-----|---------------------|--------------------|----------------------------------------|----------------|-----------------------------|--------------------|
| 26  | $N_{S,K}$           | <b>N</b>           | stoichiometric matrix                  | constant       |                             |                    |
| 197 | $E^a_K$             | <b>Ea</b>          | Arrhenius activation energy            | constant       | $kg\ m^2\ mol^{-1}\ s^{-2}$ |                    |
| 198 | $K^o_K$             | <b>Ko</b>          | Arrhenius frequency factor             | constant       | $m^{-3}\ mol\ s^{-1}$       |                    |
| 167 | $T_{N,p}$           | <b>T</b>           | link variable T to interface reactions | effort         | $K$                         | <a href="#">60</a> |
| 163 | $c_{N,S,p}$         | <b>c</b>           | link variable c to interface reactions | secondaryState | $m^{-3}\ mol$               | <a href="#">56</a> |
| 165 | $x_{N,S,p}$         | <b>x</b>           | link variable x to interface reactions | secondaryState |                             | <a href="#">58</a> |
| 168 | $f_{N,S,K,p}$       | <b>factor</b>      | factor for probability computation     | conversion     |                             | <a href="#">61</a> |
| 199 | $K_{N,K,p}$         | <b>K</b>           | Arrhenius reaction "constant"          | conversion     | $m^{-3}\ mol\ s^{-1}$       | <a href="#">94</a> |
| 169 | $\xi_{N,K,p}$       | <b>probability</b> | probability of reaction to take place  | conversion     |                             | <a href="#">62</a> |
| 200 | $\tilde{n}_{N,S,q}$ | <b>np</b>          | production from reaction set           | conversion     | $m^{-3}\ mol\ s^{-1}$       | <a href="#">95</a> |

## 6 control

|     | var              | symbol | documentation                           | type      | units | eqs |
|-----|------------------|--------|-----------------------------------------|-----------|-------|-----|
| 245 | $T_{meas_{N,p}}$ | T_meas | link variable mv I to interface control | algebraic |       | 144 |

## 7 reactions–macroscopic

|     | var          | symbol     | documentation                                                           | type | units               | eqs |
|-----|--------------|------------|-------------------------------------------------------------------------|------|---------------------|-----|
| 201 | $\_np_{I,S}$ | <b>_np</b> | link variable np to interface reactions »> macroscopic with source:node | get  | $m^{-3} mol s^{-1}$ | 96  |

## 8 macroscopic-reactions

|     | var  | symbol | documentation                                                          | type | units        | eqs |
|-----|------|--------|------------------------------------------------------------------------|------|--------------|-----|
| 162 | $_c$ | $_c$   | link variable c to interface macroscopic »> reactions with source:node | get  | $m^{-3} mol$ | 55  |
| 166 | $_T$ | $_T$   | link variable T to interface macroscopic »> reactions with source:node | get  | $K$          | 59  |
| 164 | $_x$ | $_x$   | link variable x to interface macroscopic »> reactions with source:node | get  |              | 57  |

## 9 macroscopic-control

|     | var                      | symbol             | documentation                                                           | type | units | eqs |
|-----|--------------------------|--------------------|-------------------------------------------------------------------------|------|-------|-----|
| 244 | <i>_mv_I<sub>I</sub></i> | <code>_mv_I</code> | link variable mv I to interface macroscopic »> control with source:node | get  |       | 143 |



## 10 Equations

## 11 Generic

| no | equation                                             | documentation            | layer       |
|----|------------------------------------------------------|--------------------------|-------------|
| 1  | $0 := \text{Instantiate}(\#, \#)$                    | numerical value zero     | root        |
| 2  | $1 := \text{Instantiate}(\#, \#)$                    | numerical value one      | root        |
| 3  | $0.5 := \text{Instantiate}(\#, \#)$                  | numerical value one half | root        |
| 4  | $t^o := \text{Instantiate}(t, \#)$                   | starting time            | root        |
| 5  | $t^e := \text{Instantiate}(t, \#)$                   | end time                 | root        |
| 6  | $\Delta t := \text{Instantiate}(t, \#)$              | time interval            | root        |
| 7  | $V_N := r_{xN} \cdot r_{yN} \cdot r_{zN}$            | volume                   | physical    |
| 8  | $p_N := \frac{\partial U_N}{\partial V_N}$           | thermodynamic pressure   | physical    |
| 9  | $T_N := \frac{\partial U_N}{\partial S_N}$           | temperature              | macroscopic |
| 10 | $\mu_{N,S} := \frac{\partial U_N}{\partial n_{N,S}}$ | chemical potential       | macroscopic |
| 11 | $H_N := U_N - p_N \cdot V_N$                         | Enthalpy                 | macroscopic |
| 12 | $A_N := U_N - T_N \cdot S_N$                         | Helmholtz energy         | macroscopic |
| 13 | $G_N := U_N + p_N \cdot V_N - T_N \cdot S_N$         | Gibbs free energy        | macroscopic |
| 14 | $v_{xN} := \frac{\partial r_{xN}}{\partial t}$       | velocity in x-direction  | macroscopic |
| 15 | $v_{yN} := \frac{\partial r_{yN}}{\partial t}$       | velocity in y-direction  | macroscopic |

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| no | equation                                                           | documentation                               | layer       |
|----|--------------------------------------------------------------------|---------------------------------------------|-------------|
| 16 | $v_{zN} := \frac{\partial r_{zN}}{\partial t}$                     | velocity in z-direction                     | macroscopic |
| 17 | $R := N^A \cdot k^B$                                               | gas constant                                | physical    |
| 18 | $C_{pN} := \frac{\partial H_N}{\partial T_N}$                      | total heat capacity at constant pressure    | macroscopic |
| 19 | $C_{VN} := \frac{\partial U_N}{\partial T_N}$                      | total heat capacity at constant volume      | macroscopic |
| 29 | $h_{N,S} := H_N \cdot (n_{N,S})^{-1}$                              | partial molar enthalpies                    | macroscopic |
| 30 | $m_N := \lambda_S \star^S n_{N,S}$                                 | mass                                        | macroscopic |
| 31 | $c_{N,S} := (V_N)^{-1} \cdot n_{N,S}$                              | molar concentration                         | macroscopic |
| 32 | $n_N^t := \text{reduceSum}(n_{N,S}, S)$                            | total number of moles                       | macroscopic |
| 33 | $x_{N,S} := (n_N^t)^{-1} \cdot n_{N,S}$                            | mole fraction                               | macroscopic |
| 34 | $c_{pN} := C_{pN} \cdot (m_N)^{-1}$                                | specific heat capacity at constant pressure | physical    |
| 35 | $c_{VN} := C_{VN} \cdot (m_N)^{-1}$                                | specific heat capacity at constant volume   | macroscopic |
| 36 | $\rho_N := (V_N)^{-1} \cdot m_N$                                   | density                                     | physical    |
| 40 | $A_{xyN} := r_{xN} \cdot r_{yN}$                                   | cross sectional area xy                     | physical    |
| 41 | $A_{xzN} := r_{xN} \cdot r_{zN}$                                   | cross sectional area xz                     | physical    |
| 42 | $A_{yzN} := r_{yN} \cdot r_{zN}$                                   | cross sectional area yz                     | physical    |
| 43 | $\hat{q}_{xA} := k_{xA}^q \cdot A_{yzN} \cdot F_{N,A} \star^N T_N$ | heat flow in x-direction                    | macroscopic |
| 44 | $\hat{q}_{yA} := k_{yA}^q \cdot A_{xzN} \cdot F_{N,A} \star^N T_N$ | heat flow in y-direction                    | macroscopic |

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| no | equation                                                                                                                                                                  | documentation                                        | layer       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------|
| 45 | $\hat{q}_{zA} := k_{zA}^q \cdot A_{xyN} \cdot F_{N,A} \star^N T_N$                                                                                                        | heat flow in z-direction                             | macroscopic |
| 46 | $\hat{n}_{xA,S}^d := \hat{k}_x^{d,Fick}{}_{A,S} \cdot A_{yzN} \cdot F_{N,A} \star^N c_{N,S}$                                                                              | Fick diffusion flow in x-direction                   | macroscopic |
| 47 | $\hat{n}_{yA,S}^d := \hat{k}_y^{d,Fick}{}_{A,S} \cdot A_{xzN} \cdot F_{N,A} \star^N c_{N,S}$                                                                              | Fick diffusion flow in y-direction                   | macroscopic |
| 48 | $\hat{n}_{zA,S}^d := \hat{k}_z^{d,Fick}{}_{A,S} \cdot (A_{xyN} \cdot F_{N,A}) \star^N c_{N,S}$                                                                            | Fick diffusion flow in z-direction                   | macroscopic |
| 49 | $d_A := \mathbf{sign} \left( F_{N,A} \star^N p_N \right)$                                                                                                                 | flow direction of convective flow                    | macroscopic |
| 50 | $c_{A,S} := (0.5 \cdot (F_{N,A} - d_A \cdot  F_{N,A} )) \star^N c_{N,S}$                                                                                                  | concentration in convective event-dynamic flow       | macroscopic |
| 51 | $\hat{V}_A := (\rho_A)^{-1} \cdot k_{xA}^c \cdot A_{yzN} \cdot F_{N,A} \star^N p_N$                                                                                       | volumetric flow in x-direction                       | macroscopic |
| 52 | $\hat{n}_{xA,S}^c := \hat{V}_A \cdot c_{A,S}$                                                                                                                             | molar convective flow in x-direction                 | macroscopic |
| 53 | $\mu_{N,S}^o := \mathbf{Instantiate}(\mu_{N,S}, \#)$                                                                                                                      | instantiating standard chemical potential            | macroscopic |
| 54 | $\mu_{N,S} := \mu_{N,S}^o + R \cdot T_N \cdot \mathbf{ln}(x_{N,S})$                                                                                                       | chemical potential standard model with mole fraction | macroscopic |
| 61 | $f_{N,S,K,p} := x_{N,S,p}^{(( N_S, \kappa ))}$                                                                                                                            | factor for probability computation                   | reactions   |
| 62 | $\xi_{N,K,p} := \prod_S f_{N,S,K,p}$                                                                                                                                      | probability of reaction to take place                | reactions   |
| 73 | $k_{xA,S}^d := I_{N,A} \star^N \left( (\mu_{N,S})^{-1} \cdot \left( v_{xN} \cdot \left( (V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$ | diffusivity in arc and x-direction                   | macroscopic |
| 74 | $k_{yA,S}^d := I_{N,A} \star^N \left( (\mu_{N,S})^{-1} \cdot \left( v_{yN} \cdot \left( (V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$ | diffusivity in arc and y-direction                   | macroscopic |
| 75 | $k_{zA,S}^d := I_{N,A} \star^N \left( (\mu_{N,S})^{-1} \cdot \left( v_{zN} \cdot \left( (V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$ | diffusivity in arc and z-direction                   | macroscopic |
| 76 | $k_{xA}^c := I_{N,A} \star^N \left( \left( \lambda_S \star^S (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{xN} \right)$      | convective mass conductivity in arc and x direction  | macroscopic |

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| no | equation                                                                                                                                                                                     | documentation                                       | layer       |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|
| 77 | $k_{yA}^c := I_{N,A} \stackrel{N}{\star} \left( \left( \lambda_S \stackrel{S}{\star} (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{yN} \right)$ | convective mass conductivity in arc and y-direction | macroscopic |
| 78 | $k_{zA}^c := I_{N,A} \stackrel{N}{\star} \left( \left( \lambda_S \stackrel{S}{\star} (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{zN} \right)$ | convective mass conductivity in arc and z-direction | macroscopic |
| 79 | $k_{xA}^q := I_{N,A} \stackrel{N}{\star} \left( (V_N)^{-1} \cdot C_{pN} \cdot v_{xN} \right)$                                                                                                | thermal conductivity in arc and x-direction         | macroscopic |
| 80 | $k_{yA}^q := I_{N,A} \stackrel{N}{\star} \left( (V_N)^{-1} \cdot C_{pN} \cdot v_{yN} \right)$                                                                                                | thermal conductivity in arc and y-direction         | macroscopic |
| 81 | $k_{zA}^q := I_{N,A} \stackrel{N}{\star} \left( (V_N)^{-1} \cdot C_{pN} \cdot v_{zN} \right)$                                                                                                | thermal conductivity in arc and z-direction         | macroscopic |
| 82 | $\rho_A := I_{N,A} \stackrel{N}{\star} \rho_N$                                                                                                                                               | density in arc                                      | macroscopic |
| 83 | $\hat{k}_{xA,S}^{d,Fick} := I_{N,A} \stackrel{N}{\star} \left( v_{xN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$                                            | Fick's diffusivity in arc and x-direction           | macroscopic |
| 84 | $\hat{k}_{yA,S}^{d,Fick} := I_{N,A} \stackrel{N}{\star} \left( v_{yN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$                                            | Fick diffusivity in arc and y-direction             | macroscopic |
| 85 | $\hat{k}_{zA,S}^{d,Fick} := I_{N,A} \stackrel{N}{\star} \left( v_{zN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$                                            | Fick diffusivity in arc and z-direction             | macroscopic |
| 86 | $h_{A,S} := I_{N,A} \stackrel{N}{\star} h_{N,S}$                                                                                                                                             | partial molar enthalpies in arc                     | macroscopic |
| 87 | $\dot{n}_{xN,S}^c := F_{N,A} \stackrel{A}{\star} \hat{n}_{xA,S}^c$                                                                                                                           | accumulation of molar mass due to convection        | macroscopic |
| 88 | $\dot{n}_{xN,S}^d := F_{N,A} \stackrel{A}{\star} \hat{n}_{xA,S}^d$                                                                                                                           | accumulation due to diffusion in x-direction        | macroscopic |
| 89 | $\hat{n}_{xA,S}^d := k_{xA,S}^d \cdot (A_{yzN} \cdot F_{N,A}) \stackrel{N}{\star} \mu_{N,S}$                                                                                                 | Fick diffusion flow in x-direction                  | macroscopic |
| 90 | $\hat{n}_{yA,S}^d := k_{yA,S}^d \cdot (A_{yzN} \cdot F_{N,A}) \stackrel{N}{\star} \mu_{N,S}$                                                                                                 | Fick diffusion flow in y-direction                  | macroscopic |
| 91 | $\hat{n}_{zA,S}^d := k_{zA,S}^d \cdot (A_{xyn} \cdot F_{N,A}) \stackrel{N}{\star} \mu_{N,S}$                                                                                                 | mass diffusion flow in z-direction                  | macroscopic |

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| no  | equation                                                                                                                      | documentation                                                        | layer       |
|-----|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------|
| 92  | $\dot{n}_{N,S} := \dot{n}_{xN,S}^c + \dot{n}_{xN,S}^d + V_N \cdot \tilde{n}_{N,S}$                                            | differential mass balance without reaction                           | macroscopic |
| 93  | $n_{N,S} := \int_{t^o}^{t^e} \dot{n}_{N,S} dt + n_{N,S}^o$                                                                    | fundamental state – molar mass                                       | macroscopic |
| 94  | $K_{N,K,p} := K_K^o \cdot \mathbf{exp} \left( (-E_K^a) \cdot (R \cdot T_{N,p})^{-1} \right)$                                  | Arrhenius reaction "constant"                                        | reactions   |
| 95  | $\tilde{n}_{N,S,q} := A_{N,p,q} \stackrel{p}{\star} \left( N_{S,K} \stackrel{K}{\star} (K_{N,K,p} \cdot \xi_{N,K,p}) \right)$ | production from reaction set                                         | reactions   |
| 98  | $n_{N,S}^o := \mathbf{Instantiate}(n_{N,S}, \#)$                                                                              | initial mass                                                         | macroscopic |
| 99  | $\dot{H}_{xN}^c := F_{N,A} \stackrel{A}{\star} \left( \hat{n}_{xA,S}^c \stackrel{S}{\star} h_{N,S} \right)$                   | enthalpy accumulation due to convective flow in x-direction          | macroscopic |
| 100 | $\dot{H}_{xN}^d := F_{N,A} \stackrel{A}{\star} \left( \hat{n}_{xA,S}^d \stackrel{S}{\star} h_{N,S} \right)$                   | accumulation of enthalpy due to diffusional mass flow in x-direction | macroscopic |
| 101 | $\dot{H}_{yN}^d := F_{N,A} \stackrel{A}{\star} \left( \hat{n}_{yA,S}^d \stackrel{S}{\star} h_{N,S} \right)$                   | accumulation of enthalpy due to diffusional mass flow in y-direction | macroscopic |
| 102 | $\dot{H}_{zN}^d := F_{N,A} \stackrel{A}{\star} \left( \hat{n}_{zA,S}^d \stackrel{S}{\star} h_{N,S} \right)$                   | accumulation of enthalpy due to diffusional mass flow in z-direction | macroscopic |
| 103 | $\dot{q}_{xN} := F_{N,A} \stackrel{A}{\star} \hat{q}_{xA}$                                                                    | accumulation due to heat flow in x-direction                         | macroscopic |
| 104 | $\dot{q}_{yN} := F_{N,A} \stackrel{A}{\star} \hat{q}_{yA}$                                                                    | accumulation due to heat flow in y-direction                         | macroscopic |
| 105 | $\dot{q}_{zN} := F_{N,A} \stackrel{A}{\star} \hat{q}_{zA}$                                                                    | accumulation due to heat flow in z-direction                         | macroscopic |
| 106 | $\hat{w}_A := \mathbf{Instantiate}(\hat{q}_{xA}, \#)$                                                                         | a fixed work flow to start with                                      | macroscopic |
| 107 | $\dot{n}_{yN,S}^d := F_{N,A} \stackrel{A}{\star} \hat{n}_{yA,S}^d$                                                            | accumulation due to diffusion in y-direction                         | macroscopic |

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| no  | equation                                                                                                                                  | documentation                                        | layer       |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------|
| 108 | $\dot{n}_{zN,S}^d := F_{N,A} \overset{A}{\star} \hat{n}_{zA,S}^d$                                                                         | accumulation due to diffusion in z-direction         | macroscopic |
| 109 | $\dot{w}_N := F_{N,A} \overset{A}{\star} \dot{w}_A$                                                                                       | accumulation of enthalpy due to work flow            | macroscopic |
| 110 | $\dot{H}_N := \dot{H}_{xN}^c + \dot{H}_{xN}^d + \dot{H}_{yN}^d + \dot{H}_{zN}^d + \dot{q}_{xN} + \dot{q}_{yN} + \dot{q}_{zN} + \dot{w}_N$ | accumulation of enthalpy                             | macroscopic |
| 111 | $H_N^o := \mathbf{Instantiate}(H_N, \#)$                                                                                                  | initial enthalpy                                     | macroscopic |
| 112 | $H_N := \int_{t^o}^{t^e} \dot{H}_N dt + H_N^o$                                                                                            | Enthalpy                                             | macroscopic |
| 113 | $U_N^e := (C_N)^{-1} \cdot U_N$                                                                                                           | electrical potential – voltage                       | macroscopic |
| 114 | $I_N^e := \frac{dC_N}{dt}$                                                                                                                | current definition                                   | macroscopic |
| 115 | $R_N^e := (I_N^e)^{-1} \cdot U_N^e$                                                                                                       | electrical resistant                                 | macroscopic |
| 116 | $\dot{U}_A^e := F_{N,A} \overset{A}{\star} (R_N^e \cdot I_N^e)$                                                                           | Kirkhoffs first law                                  | macroscopic |
| 117 | $C_{pN} := m_N \cdot c_{pN}$                                                                                                              | total heat capacity at constant pressure             | macroscopic |
| 119 | $T_N^{ref} := \mathbf{Instantiate}(T_N, \#)$                                                                                              | reference temperature                                | macroscopic |
| 120 | $c_{pN} := \mathbf{Instantiate}(c_{pN}, \#)$                                                                                              | constant specific heat capacity at constant pressure | macroscopic |
| 121 | $T_N := H_N \cdot (C_{pN})^{-1} + T_N^{ref}$                                                                                              | temperature from constant heat capacity              | macroscopic |
| 122 | $T_N^n := \mathbf{Instantiate}(T_N, \#)$                                                                                                  | value to norm measurement of temperature             | macroscopic |
| 123 | $\bar{T}_N := T_N \cdot (T_N^n)^{-1}$                                                                                                     | temperature measurement                              | macroscopic |

*Continued on next page*

| no  | equation                                  | documentation        | layer       |
|-----|-------------------------------------------|----------------------|-------------|
| 133 | $\hat{m}_{N,A} := \hat{V}_A \cdot \rho_N$ | convective mass flow | macroscopic |

## 12 Interface Link Equation

| no  | equation                                                                                                                                  | documentation      | layer                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------|
| 55  | $\_c_{I,S} := F^{source}_{N,I} \star^N c_{N,S}$                                                                                           | interface equation | macroscopic -> reactions |
| 56  | $c_{N,S,p} := (F^{sink}_{N,I} \cdot \_c_{I,S}) \star^I S_{I,p}$                                                                           | interface equation | reactions                |
| 57  | $\_x_{I,S} := F^{source}_{N,I} \star^N x_{N,S}$                                                                                           | interface equation | macroscopic -> reactions |
| 58  | $x_{N,S,p} := (F^{sink}_{N,I} \cdot \_x_{I,S}) \star^I S_{I,p}$                                                                           | interface equation | reactions                |
| 59  | $\_T_I := F^{source}_{N,I} \star^N T_N$                                                                                                   | interface equation | macroscopic -> reactions |
| 60  | $T_{N,p} := (F^{sink}_{N,I} \cdot \_T_I) \star^I S_{I,p}$                                                                                 | interface equation | reactions                |
| 96  | $\_np_{I,S} := \text{reduceSum} \left( \left( \left( F^{source}_{N,I} \star^N \tilde{n}_{N,S,q} \right) \cdot S_{I,q} \right), q \right)$ | interface equation | reactions -> macroscopic |
| 97  | $\tilde{n}_{N,S} := F^{source}_{N,I} \star^I \_np_{I,S}$                                                                                  | interface equation | macroscopic              |
| 143 | $\_mv\_I_I := F^{source}_{N,I} \star^N \bar{T}_N$                                                                                         | interface equation | macroscopic -> control   |
| 144 | $T_{meas_{N,p}} := (F^{sink}_{N,I} \cdot \_mv\_I_I) \star^I S_{I,p}$                                                                      | interface equation | control                  |